Willem Davison

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Profile

AI, Robotics, and Computer Vision PhD student at the University of Edinburgh with a 1st-class MSci in Theoretical Physics. Specialising in dynamics and motion modeling for applications in efficient video understanding and deployable embodied AI. Combining proven professional engineering experience with advanced Python and PyTorch development to build scalable, end-to-end AI architectures.

Professional Experience

University of Edinburgh, School of Informatics August 2024 - Present

PhD Researcher under Dr Laura Sevilla-Lara's supervision

- Designed and developed novel machine learning pipelines or applications in both advanced computer vision tasks and robot manipulation tasks.
- Organised and managed video computer vision research group meetings and presentations.

BAE Systems October 2023 - August 2024

Lead Electromagnetics Engineer

- Designed experiments and developed C++ applications for the statistical analysis of raw material measurements from vector network analysers.
- Implemented numerical models to infer and compare key material properties from noisy data.
- Managed and mentored junior apprentices, leading to promotion to placement manager responsible for overseeing project development, providing technical guidance, and apprentice interviews.

Research

AntVLA: Enabling Dynamic Manipulation Capability in VLAs (Ongoing)

- Modelling temporal dynamics to adapt robot actions, endowing a frozen VLA with dynamic manipulation capability while preserving its task-level competence.

It's a Matter of Time: Three Lessons on Long-Term Motion for Perception

[Preprint arXiv ID: 2602.14705] *First Author Paper, Under Review*

- Proposed MovT, a novel Transformer architecture that leverages sparse point-tracks (via CoTracker3) to outperform pixel-based baselines.
- Engineered distributed GPU HPC pipelines using Bash and PyTorch to process ∼100k videos from large video datasets, extracting low-dimensional signals from noisy raw data.

Skills

Programming Languages: Python (Advanced), C++ (Advanced), Bash (Advanced).

Python Libraries: NumPy, Pandas, PyTorch, TensorFlow, Scikit-learn, SciPy, Matplotlib.

AI, Robotics, & Computer Vision Expertise: Foundation Models, Video Understanding, Transformers, Isaac-Sim, Reinforcement Learning, Point-Tracking.

Education

University of Nottingham September 2019 – May 2023

Physics with Theoretical Physics MSci - 1st Class

- **Master's Thesis Project:** Derived supersymmetric black hole solutions to the Einstein-Maxwell field equations by exploiting advanced differential spinorial geometry methods.
- **Scholarships Awarded:** Sir Peter Mansfield, First Year Academic Excellence

Durham Johnston Comprehensive School - Sixth Form September 2017 – June 2019

A*, A*, A* (Physics, Maths, and Further Maths)

Interests

I am an avid cyclist and hiker. I've recently completed the West Highland Way and a two-week cycle trip through France, Belgium, and the Netherlands with a close friend. I also enjoy representing the School of Informatics Squash team in intramural tournaments. To exercise my creative side, I make jewellery and frequently practise photography.